

LOVETCH, Bulgaria

WMO: 155250

Lat: 43.150N

Lon: 24.700E

Elev: 221

StdP: 98.70

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.9	-9.1	-14.2	1.1	-11.3	-11.3	1.5	-8.6	16.7	0.6	13.7	3.6	1.6	180	0.455

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.0	34.6	21.9	32.7	21.5	31.1	21.0	23.5	31.7	22.5	30.4	21.7	29.3	2.4	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.8	15.9	28.0	19.9	15.0	26.6	19.0	14.2	25.7	71.2	32.0	67.3	30.5	64.2	29.5	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.6	8.2	6.3	-14.9	37.3	3.1	2.4	-17.2	39.0	-19.0	40.5	-20.7	41.8	-23.0	43.6		
(5)			-15.3	25.3	3.1	1.2	-17.5	26.1	-19.3	26.9	-21.0	27.5	-23.2	28.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.5	0.5	3.0	7.6	12.5	17.3	21.1	23.3	23.4	18.8	12.8	7.3	2.0
	DBStd	9.09	5.59	5.87	4.97	3.86	3.29	3.18	2.82	2.91	3.80	4.50	4.81	5.10
	HDD10.0	1007	296	201	107	19	1	0	0	0	0	24	108	251
	HDD18.3	2619	552	428	333	178	59	11	2	2	41	177	331	507
	CDD10.0	1925	2	6	32	94	226	332	413	416	263	112	28	3
	CDD18.3	498	0	0	0	3	26	92	156	160	53	6	1	0
	CDH23.3	5056	0	0	6	49	272	905	1586	1648	523	66	3	0
	CDH26.7	2046	0	0	1	8	51	334	699	759	182	12	0	0
Wind	WSAvg	2.5	2.4	2.7	3.1	2.8	2.4	2.5	2.6	2.3	2.4	2.3	2.3	2.3
Precipitation	PrecAvg	664	41	38	49	56	77	74	80	51	62	50	37	49
	PrecMax	1076	122	71	101	128	154	191	228	147	191	103	105	92
	PrecMin	409	10	10	12	6	21	9	15	5	2	2	4	9
	PrecStd	158	25	17	21	28	34	46	53	39	47	28	22	23
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.4	19.6	23.8	27.7	30.5	34.7	38.0	37.0	33.9	28.2	23.0	17.5
		MCWB	10.0	11.8	13.2	17.0	20.1	22.6	21.5	21.6	20.2	18.4	14.8	11.0
	2%	DB	13.0	16.3	20.8	24.5	28.0	32.1	34.5	34.8	30.9	25.1	20.2	13.9
		MCWB	8.3	10.0	12.2	15.8	19.1	22.2	21.9	22.0	19.3	16.6	13.4	9.2
	5%	DB	10.5	13.9	18.3	21.7	26.3	30.3	32.5	32.7	28.6	22.4	17.2	11.2
		MCWB	7.0	8.8	10.9	14.3	18.4	21.6	21.7	21.4	18.6	15.6	12.2	7.7
	10%	DB	8.1	11.3	15.7	19.6	24.6	28.5	30.5	31.0	26.4	20.0	14.3	8.7
		MCWB	5.5	7.5	9.7	13.1	17.5	20.6	21.3	20.7	17.7	14.4	10.7	6.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.4	12.0	14.1	18.1	21.4	24.6	24.5	24.7	21.1	18.9	15.3	11.4
		MCDB	15.3	18.3	21.5	25.8	28.6	32.0	33.4	32.7	30.9	26.9	21.7	16.0
	2%	WB	8.6	10.4	12.5	16.3	20.0	23.2	23.3	23.3	20.0	17.3	13.8	9.4
		MCDB	12.4	15.7	20.0	23.5	26.7	30.6	31.7	31.8	28.8	23.6	19.3	13.4
	5%	WB	7.2	9.0	11.4	14.8	18.9	22.1	22.4	22.1	19.1	16.1	12.5	8.0
		MCDB	10.2	13.6	17.7	20.4	25.1	29.2	30.3	30.7	27.1	21.4	16.5	10.9
	10%	WB	5.6	7.6	10.0	13.7	17.9	21.0	21.6	21.2	18.2	14.9	11.1	6.4
		MCDB	8.0	10.9	15.4	18.5	23.7	27.3	29.1	29.2	25.1	19.4	14.0	8.5
Mean Daily Temperature Range	5% DB	MDBR	6.9	8.0	9.3	10.2	10.9	11.3	12.0	12.6	11.3	9.6	7.6	6.7
		MCDBR	11.0	12.8	13.8	14.1	13.9	14.5	15.5	15.8	15.4	14.0	12.4	11.1
		MCWBR	7.3	7.8	7.5	7.4	7.0	6.7	6.4	6.1	6.4	6.8	7.0	7.4
	5% WB	MCDBR	10.8	12.1	13.0	12.6	12.8	13.3	13.5	13.8	13.8	12.7	11.0	10.3
		MCWBR	7.3	7.5	7.3	6.9	6.8	6.6	6.2	6.0	6.1	6.5	6.4	7.0
Clear-Sky Solar Irradiance	taub	0.339	0.371	0.416	0.475	0.454	0.457	0.465	0.475	0.444	0.400	0.364	0.342	
	taud	2.297	2.220	2.146	1.996	2.125	2.159	2.146	2.112	2.167	2.270	2.335	2.314	
	Ebn at Noon	774	808	815	793	824	821	809	785	778	765	739	732	
	Edh at Noon	86	110	135	169	153	148	148	149	131	104	84	78	
All-Sky Solar Radiation	RadAvg	1.52	2.34	3.45	4.67	5.72	6.31	6.45	5.79	4.25	2.79	1.67	1.27	
	RadStd	0.23	0.30	0.33	0.49	0.47	0.43	0.42	0.40	0.33	0.33	0.21	0.17	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.46	N/A	N/A	N/A	+0.45	N/A	N/A	N/A	N/A	N/A
(47)	Regional (47 neighbors)	+0.44	N/A	N/A	N/A	+0.50	+0.48	N/A	-138	N/A	N/A

Nomenclature: See separate page